

# PLANRS ACADEMY MASTER COURSE

Grade 4 | Graduation Showcase — The Junior Innovator's Guide to Design Thinking

## Welcome to the World of Design!

Did you know that you are already a designer? Every time you find a way to make something work better, you are using Design Thinking.

**What is a Gadget?** A gadget is more than just a piece of technology. It is a tool that solves a problem for people. It can be as simple as a tool to pump air or as complex as a computer.

**Your Mission Today:** In this lesson, you will learn how real heroes use simple materials to solve big problems, and you will start your own journey using the Design Loop.

**Mini-Task:** Look around right now. Spot 3 gadgets you use every day. Think about your water bottle, cycle pump, or lunchbox.

## 1. Functional Design

Design is not just about how things look—it's about how they work! We call this **Functional Design**: design made to serve a special purpose.

- **The Idli Stand:** A brilliant design that saves space by stacking plates on top of each other.
- **The Cycle Pump:** A functional tool that keeps your bike ready to ride whenever you need it.

## 2. Meet Real-Life Heroes

Meet two amazing Indian innovators who turned simple ideas into gadgets that changed lives:

### Mansukhbhai Prajapati (The Clay Innovator)

He created **Mitticool**, a refrigerator made completely out of clay! It keeps water, milk, and vegetables fresh and cool without using any electricity.

### Remya Jose (The Bicycle Laundry Star)

When she was a school student, she invented a washing machine attached to a bicycle! By pedalling the cycle, she could wash clothes while exercising, saving both electricity and time.

## 3. The Design Loop: THINK

How do inventors come up with great ideas? They use a map called the **Design Loop**! The loop has three main steps that keep going around: **THINK**, **MAKE**, and **SHARE**.

## STEP 1: THINK - Find the Problem

Every great gadget starts by looking around and finding a problem to solve. Design thinkers are always asking questions like:

- What is hard to do?
- What breaks often?
- How can we make this task more fun or faster?

### Brainstorming Challenge:

Look at this list of daily hassles. Which one would you like to build a gadget for?

- The water bottle is too heavy to carry.
- Pencils keep rolling off the desk and losing their tips.
- The tiffin box leaks oil into the school bag.

## 4. The Design Loop: MAKE

### STEP 2: MAKE - Build and Test

This is where you become a maker! You use simple materials to create a **Prototype**—a rough model of your gadget.

#### The Materials Station:

You don't need expensive factory tools. You can make an amazing prototype using things found around your house or classroom:

- ✓ Cardboard boxes, plastic cups, and bottles
- ✓ Rubber bands, popsicle sticks, and straws
- ✓ Paper clips, string, tape, and clay

*[Visual Idea: A blueprint drawing setup of cardboard pieces and rubber bands showing what your prototype might look like.]*

## 5. The Design Loop: SHARE

### STEP 3: SHARE - Tell Your Story

Show your gadget to your friends, family, and classmates. Tell them how it helps solve the problem you found.

#### Why Sharing Matters:

- **Get Better Ideas:** When others see your gadget, they might have great suggestions to improve it.
- **Inspire Others:** Your invention could inspire someone else to create something amazing too!

### Sharing Challenge:

Prepare a "30-second pitch" for your gadget. Explain:

1. What the problem was.
2. How your gadget solves it.
3. One thing you are proud of!

## 6. 3-2-1 Exit Review

Let's check what we've learned today to make sure you're ready to be an inventor!

- **3 Gadgets:** List 3 gadgets you learned about today (from the lesson or your surroundings).
- **2 Steps:** Write down 2 steps of the Design Loop that you remember.
- **1 Feeling Word:** What is one word that describes how you feel about being an innovator?

### Your Final Challenge!

Find one problem at home or school. Use the Design Loop to THINK, MAKE, and SHARE your gadget! Bring your idea or prototype to the graduation showcase.

### Future Innovator Ready!

Plants Academy | Grade 4 Demo Day | Graduation Showcase Series

# Practice Worksheet: Future Innovators Challenge

Show what you know about Design Thinking! Answer the questions below using what you learned.

## Question 1

What is a **gadget**? Give one simple example mentioned in the text.

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## Question 2

Explain what **Functional Design** means in your own words.

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## Question 3

How does the **Idli Stand** show good functional design? What problem does it solve?

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## Question 4

What amazing clay gadget did **Mansukhbhai Prajapati** create, and why is it special?

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## Question 5

How did **Remya Jose** find a way to wash clothes and exercise at the same time?

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### Question 6

Name the three main steps of the **Design Loop**.

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### Question 7

What questions do design thinkers ask when they are in the **THINK** stage?

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### Question 8

What is a **Prototype**, and during which step of the Design Loop do you build it?

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### Question 9

List four simple everyday materials you can use to build a prototype at home.

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### Question 10

Why is it important to **SHARE** your invention with friends and family? Give two reasons.

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### Question 11

Imagine your pencil keeps rolling off your desk. Write down one quick idea to fix this problem.

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### Question 12

What are the three things you should explain when preparing a **30-second pitch**?

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